Date :.....

BRACU Mongol Toirl-(Recruitment)

Rover Autonomy Recruitment project

Task1: (Route Planning)

Given Information:

Latitude = 38.418012°

Longitude = −110.777203°

Waypoint 1 (WP1):

Latitude = 38.418603°

Longitude = −110.775784°

Waypoint 2 (WP2):

Latitude = 38.420167°

Longitude = −110.776319°

Waypoint 3 (WP3):

Latitude = 38.420289°

Longitude = −110.776246°

Initial Rover Heading = 0° (North).

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Distance Calculation Using Haversine Formula

Start → WP1 distance

$$a = \sin^2\left(\frac{\Delta\phi}{2}\right) + \cos\phi_1 \cos\phi_2 \sin^2\left(\frac{\Delta\lambda}{2}\right)$$

$$d = 2R\sin^{-1}\left(\sqrt{a}\right)$$

$\therefore \Delta\phi = \phi_2 - \phi_1 = 0.000591°$

$\therefore \Delta\lambda = \lambda_1 - \lambda_2 = 0.001419°$

Degrees to radians:

$$\Delta\phi = 0.000591° \times \frac{\pi}{180}$$

$$\Delta\lambda = 0.001419° \times \frac{\pi}{180}$$

$\therefore d = 140.01$ meters

WP1 → WP2 distance:

$\therefore \Delta\phi = \phi_2 - \phi_2 = 0.001569°$

$\therefore \Delta\lambda = \lambda_2 - \lambda_1 = -0.000535°$

where,

$R = 6,371,000$ meters (Earth's radius)

ϕ = Latitude in radius

λ = longitude in radius

d = Distance in meters

Given,

$\phi_1 = 38.918012°$

$\phi_2 = 38.918603°$

$\lambda_1 = -110.777203°$

$\lambda_2 = -110.775784°$

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degrees to radians:

$\Delta\varphi = 0.001564^\circ \times \frac{\pi}{180}$

$\Delta\lambda = -0.000535 \times \frac{\pi}{180}$

$d = 180.05$ meters

Given,

$\varphi_1 = 38.418603^\circ$

$\varphi_2 = 38.420167^\circ$

$\lambda_1 = -110.775784^\circ$

$\lambda_2 = -110.776319^\circ$

wp2 → wp3

$\Delta\varphi = \varphi_2 - \varphi_1 = 0.000122^\circ$

$\Delta\lambda = 0.000073 \times \frac{\pi}{180}$

$\therefore d = 14.98$ meters.

Given,

$\varphi_1 = 38.420167^\circ$

$\varphi_2 = 38.420289^\circ$

$\lambda_1 = -110.776319^\circ$

$\lambda_2 = -110.776296^\circ$

degrees to radians:

$\Delta\varphi = 0.000122 \times \frac{\pi}{180}$

$\Delta\lambda = 0.000073 \times \frac{\pi}{180}$

$\therefore d = 14.98$ meters.

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degrees to radians:

$\Delta\varphi = 0.001564^\circ \times \frac{\pi}{180}$

$\Delta\lambda = -0.000535 \times \frac{\pi}{180}$

$d = 180.05$ meters

Given,

$\varphi_1 = 38.918603^\circ$

$\varphi_2 = 38.920167^\circ$

$\lambda_1 = -110.775784^\circ$

$\lambda_2 = -110.776319^\circ$

WP2 → WP3

$\Delta\varphi = \varphi_2 - \varphi_1 = 0.000122^\circ$

$\Delta\lambda = 0.000073 \times \frac{\pi}{180}$

$\therefore d = 14.98$ meters.

Given,

$\varphi_1 = 38.920167^\circ$

$\varphi_2 = 38.920289^\circ$

$\lambda_1 = -110.776319^\circ$

$\lambda_2 = -110.776296^\circ$

degrees to radians:

$\Delta\varphi = 0.000122 \times \frac{\pi}{180}$

$\Delta\lambda = 0.000073 \times \frac{\pi}{180}$

$\therefore d = 14.98$ meters.

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Bearing and Turning Instructions

(1) Bearing Calculations

Bearing is measured clockwise from North(0°). The initial heading of the rover is North (0°) Using the initial heading and the calculated bearings:

- Start → wp1 = 62.01°
- wp1 → wp2 = 345.00°
- wp2 → wp3 = 25.12°

(2) Relative Turning Angles:

Leg 1: Start → wp1

Initial heading = 0°

New bearing = 62.01°

Turn = 62.01° RIGHT

Leg 2: wp1 → wp2

Previous heading = 62.01°

New bearing = 345.00°

Turn = 62.01° − 345.00°

= −282.99°

Taking the shorter turning direction:

Turn = 77.01° LEFT

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Leg 3: WP2 → WP3

Previous heading = 395.00°

New bearing = 25.12°

Turn = 25.12° − 395.00° = −319.88°

Taking the shorter turning direction:

Turn = 40.12° RIGHT.

(3) Final Rover Command Sequence:

Step	Rover Command
1	Turn RIGHT 62.01°
2	Drive 190.01 m
3	Turn LEFT 77.01°
4	Drive 180.05 m
5	Turn RIGHT 40.12°
6	Drive 14.98 m.

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GPS Accuracy Analysis (± 3 Metres)

(1) GPS Accuracy:

The GPS receiver has an accuracy of approximately ± 3 metres. This means the actual rover position may differ from the reported GPS position by up to 3 metres.

(2) Effect on Navigation:

- The rover may not reach the exact waypoint coordinates.
- Small position errors can affect distance and bearing calculations.
- Error may become more noticeable when the rover moves between nearby waypoints.
- The rover should use a suitable arrival tolerance instead of expecting an exact coordinate match.

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- The third leg is only about 14.98 metres long. therefore, a position uncertainty of several metres can have a significant effect on its calculated direction and distance.

) Possible solutions:

- using multiple GPS readings and filtering to reduce random fluctuations.
- Recalculating the rover's position and direction regularly.
- Using an IMU to help estimate the rover's orientation and movement.
- Applying a suitable waypoint arrival tolerance to avoid unnecessary corrections caused by small GPS variations.

Conclusion :

A GPS accuracy reduction of ±3 metres can make the calculated distances and bearing less

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reliable. This may cause incorrect turns, inaccurate driving distances and accumulated navigation errors.

Therefore, GPS filtering, regular position updates and additional sensors are important for improving the rover's navigation accuracy.